PREDICTIVE FORECASTING OF HHS UNACCOMPANIED CHILDREN PROGRAM

Submitted by:

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Internship Project Report

Organization:

Unified Mentor

Domain:

Data Analytics

Tools & Technologies Used:

Python, Google Colab, Streamlit, Machine Learning, Time Series Forecasting

Submission Year:

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ABSTRACT

This project focuses on predictive forecasting analysis for the HHS Unaccompanied Children Program using historical care load and discharge data. The primary objective of this project is to analyse operational trends, identify forecasting patterns, and build predictive models for future care load estimation.

The project includes data preprocessing, exploratory data analysis (EDA), feature engineering, statistical forecasting, and machine learning-based forecasting techniques. Multiple forecasting approaches including Random Forest, Gradient Boosting, Moving Average, ARIMA, and Exponential Smoothing were implemented and evaluated using forecasting performance metrics such as MAE and RMSE.

An interactive Streamlit dashboard was also developed to visualize forecasting trends, compare forecasting models, analyse KPI metrics, and support operational decision-making processes. The dashboard includes model selection, forecast horizon analysis, confidence interval visualization, and scenario comparison features.

The project demonstrates how predictive analytics and forecasting systems can improve resource planning, operational forecasting, and data-driven decision making in real-world environments.

TABLE OF CONTENTS

INTRODUCTION

Forecasting is one of the most important applications of data analytics because it helps organizations predict future operational conditions based on historical data patterns.

The HHS Unaccompanied Children Program manages children transferred from CBP custody into HHS care facilities. Since the number of children entering and leaving care facilities changes frequently, forecasting future care load becomes essential for proper operational planning.

This project focuses on applying predictive analytics, time-series forecasting, and machine learning techniques to analyse historical operational data and forecast future care demand.

The project also demonstrates how interactive dashboards can help stakeholders visualize forecasting insights and compare prediction performance across multiple forecasting models.

PROBLEM STATEMENT

The HHS Unaccompanied Children Program experiences fluctuations in:

- care load
- discharge demand
- transfer rates
- operational resource requirements

Without accurate forecasting systems, it becomes difficult for organizations to:

- allocate resources efficiently
- prepare operational capacity
- monitor future demand
- make proactive decisions

Traditional manual forecasting methods are often insufficient for handling large operational datasets.

Therefore, this project aims to build an automated predictive forecasting solution capable of:

- identifying historical trends
- forecasting future care load
- comparing forecasting models
- visualizing insights through dashboards

The project combines both statistical forecasting and machine learning forecasting approaches for improved prediction analysis.

OBJECTIVES

The main objective of this project is to develop a forecasting analytics system for the HHS Unaccompanied Children Program.

Specific objectives include:

Data Preparation

- clean and preprocess historical operational data
- handle missing values and inconsistencies

Exploratory Analysis

- identify trends and forecasting behavior
- understand historical operational patterns

Feature Engineering

- create lag variables
- create rolling statistics
- improve forecasting model performance

Forecasting

- build multiple forecasting models
- compare statistical and machine learning approaches

Dashboard Development

- build interactive Streamlit dashboard
- visualize forecasting insights
- support operational analysis

Performance Evaluation

- evaluate forecasting accuracy using MAE and RMSE
- compare model performance

DATASET DESCRIPTION

The dataset used in this project contains historical operational records related to the HHS Unaccompanied Children Program.

Dataset Preview

	Date	Children apprehended and placed in CBP custody*	Children in CBP custody	Children transferred out of CBP custody	Children in
712	2023-01-31 00:00:00	26	36	36	
711	2023-02-01 00:00:00	25	32	27	
710	2023-02-02 00:00:00	15	13	23	
709	2023-02-05 00:00:00	16	18	15	
708	2023-02-06 00:00:00	11	22	12	
707	2023-02-07 00:00:00	16	20	20	
706	2023-02-08 00:00:00	93	196	169	
705	2023-02-09 00:00:00	124	234	161	
704	2023-02-12 00:00:00	92	203	173	
703	2023-02-13 00:00:00	186	259	172	

Figure 1: **Dataset Preview Used for Predictive Forecasting Analysis**

The dataset includes information such as:

- number of children in CBP custody
- number of children transferred out of custody
- number of children in HHS care
- number of discharged children
- historical timeline information

The dataset represents time-series operational data and is suitable for predictive forecasting analysis.

The data was analysed over historical periods to identify trends, fluctuations, and operational forecasting behavior.

This dataset helped develop forecasting models capable of predicting future care load demand.

DATA PREPROCESSING

Raw datasets often contain:

- missing values
- formatting inconsistencies
- text-based numeric values
- timeline issues

Therefore, preprocessing is one of the most important stages of the forecasting pipeline.

The following preprocessing tasks were performed:

Date Conversion

The Date column was converted into datetime format for time-series analysis.

Missing Value Handling

Rows containing missing values were identified and removed to improve forecasting reliability.

Numeric Conversion

Comma-separated numeric columns were converted into float format for machine learning compatibility.

Data Sorting

The dataset was sorted chronologically to maintain forecasting sequence integrity.

Feature Preparation

Historical lag features and rolling averages were generated for forecasting analysis.

These preprocessing steps improved data quality and forecasting model performance.

EXPLORATORY DATA ANALYSIS (EDA)

Exploratory Data Analysis (EDA) was performed to understand:

- historical trends
- operational fluctuations
- forecasting behavior
- relationships between variables

Several visualizations and analyses were conducted.

Trend Analysis

Historical trends in HHS care load were visualized using line charts.

Correlation Analysis

Correlation heatmaps helped identify relationships between operational variables.

Monthly Trend Analysis

Monthly average patterns were analysed to identify periodic fluctuations.

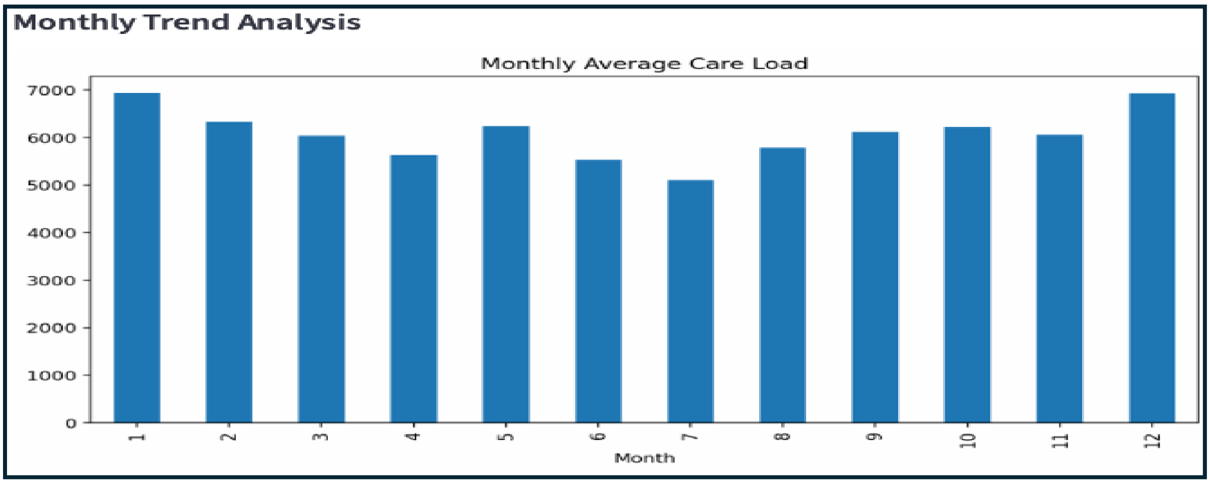

Figure 2: **Monthly Average Care Load Analysis**

Forecast Pattern Visualization

Forecast trends and operational movement patterns were studied.

EDA helped identify important forecasting characteristics and improved feature selection for forecasting models.

FEATURE ENGINEERING

Feature engineering improves forecasting performance by creating meaningful variables from historical data.

The following forecasting features were created:

Lag Features

Lag variables help forecasting models understand previous historical conditions.

Examples:

- lag_1
- lag_7

Rolling Mean Features

Rolling averages help smooth short-term fluctuations and capture long-term trends.

Feature engineering improved the predictive capability of machine learning forecasting models.

FORECASTING MODELS

Multiple forecasting models were implemented to compare forecasting performance.

Random Forest Regressor

Random Forest is an ensemble machine learning algorithm that combines multiple decision trees for prediction.

Advantages:

- strong prediction capability
- handles nonlinear patterns
- reduces overfitting

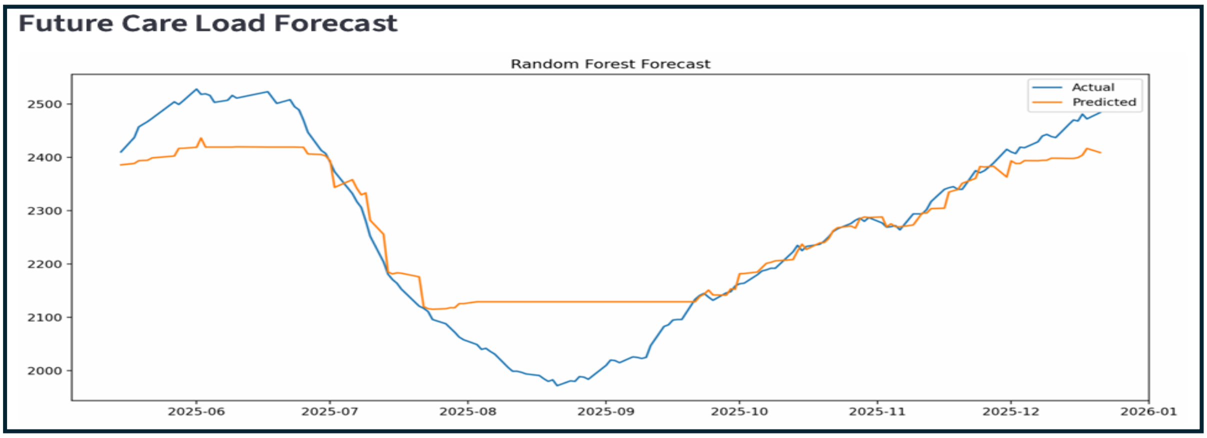

Figure 3: **Random Forest Forecasting Visualization**

Gradient Boosting Regressor

Gradient Boosting sequentially improves prediction performance by correcting forecasting errors.

Advantages:

- high forecasting accuracy
- optimized learning
- strong performance on structured data

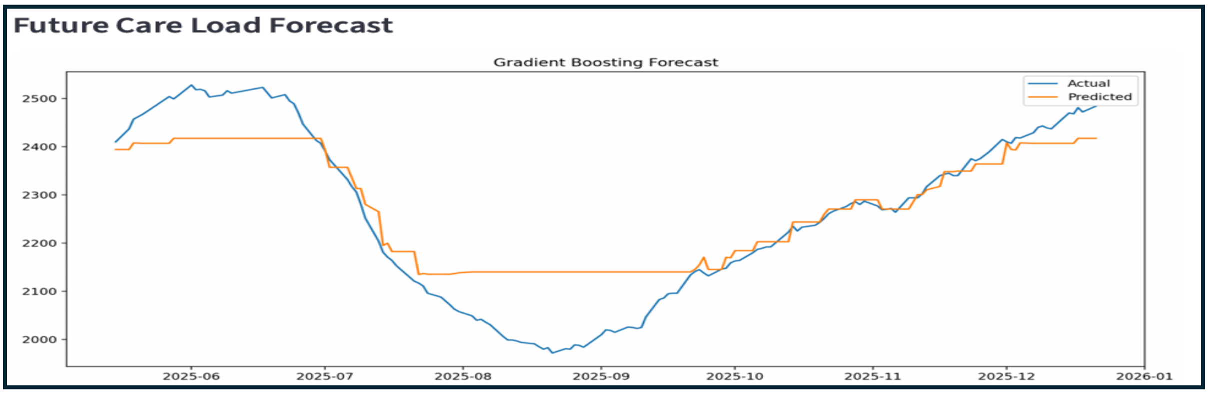

Figure 4: **Gradient Boosting Forecasting Visualization**

Moving Average Forecasting

Moving Average forecasting smooths short-term fluctuations and identifies trend movement.

Advantages:

- simple implementation
- statistical forecasting approach
- useful for trend smoothing

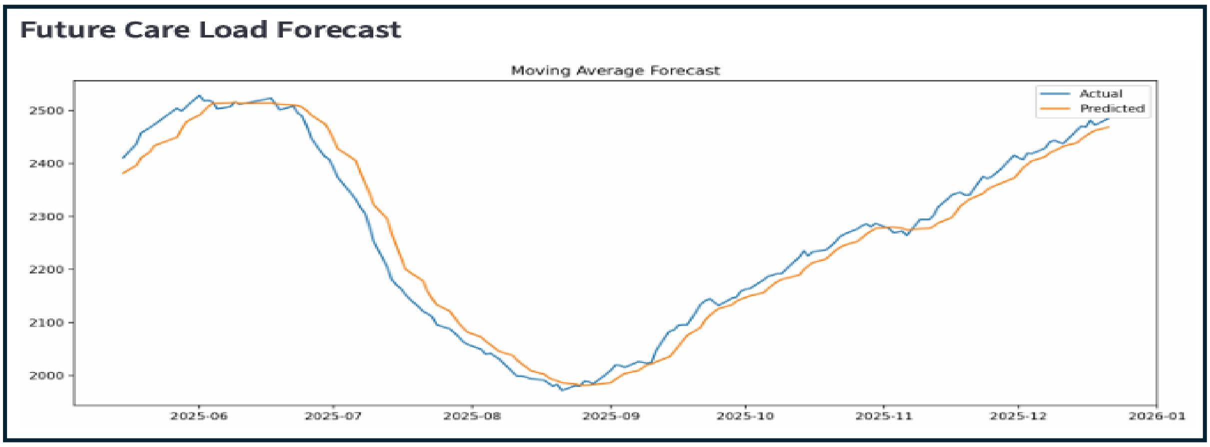

Figure 5: **Moving Average Forecasting Visualization**

ARIMA Forecasting

ARIMA is a statistical time-series forecasting model.

Advantages:

- captures temporal dependencies

- effective for sequential forecasting
- suitable for trend analysis

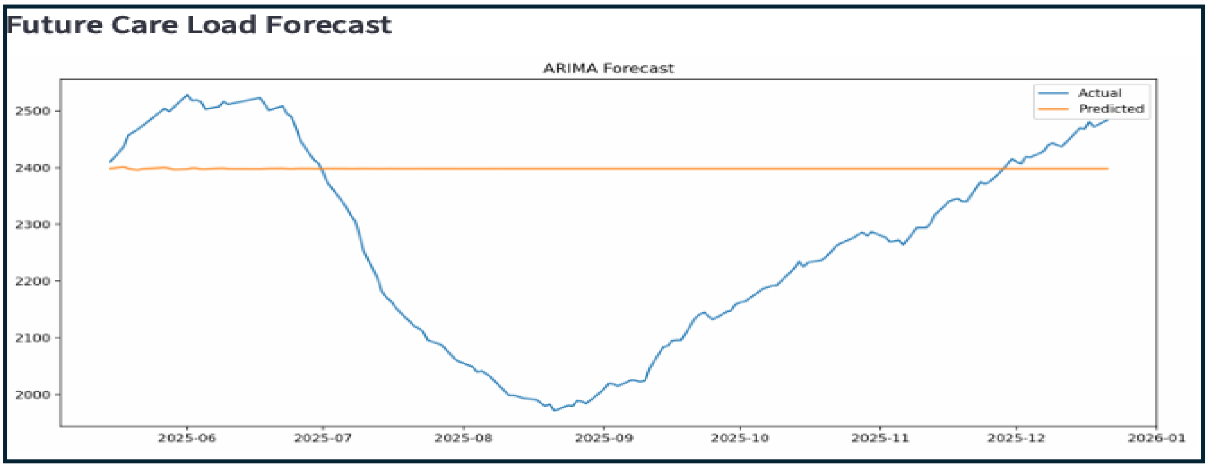

Figure 6: **ARIMA Forecasting Visualization**

<u>Exponential Smoothing</u>

Exponential Smoothing gives higher importance to recent observations.

Advantages:

- smooth forecasting
- trend analysis
- operational forecasting

EVALUATION METRICS

Forecasting models were evaluated using multiple performance metrics.

Mean Absolute Error (MAE)

MAE measures average forecasting error magnitude.

Lower MAE indicates better prediction quality.

Root Mean Squared Error (RMSE)

RMSE measures squared forecasting errors.

Lower RMSE indicates stronger forecasting performance.

Forecast Accuracy Percentage

Accuracy percentage helps compare forecasting models and evaluate operational prediction quality.

These metrics helped compare statistical and machine learning forecasting approaches.

Model Comparison Metrics

Model	MAE	RMSE
Random Forest	50.7700	68.5900
Gradient Boosting	53.5700	72.5700
Moving Average	25.0900	30.8700
ARIMA	189.5300	230.6300

Figure 7: **Forecasting Model Performance Comparison Metrics**

DASHBOARD DEVELOPMENT

An interactive Streamlit dashboard was developed to visualize forecasting insights and improve accessibility of analytical information.

The dashboard provides:

- forecasting visualization
- model comparison
- operational KPI analysis
- forecasting interaction

Dashboard Features

Forecast Horizon Selector

Allows users to select future forecasting duration.

Forecasting Model Toggle

Allows switching between forecasting models.

KPI Cards

Displays forecasting accuracy and operational metrics.

Confidence Interval Visualization

Shows uncertainty range in forecasting predictions.

Scenario Comparison Visualization

Compares forecasting outputs across multiple models.

Monthly Trend Analysis

Displays historical monthly forecasting behavior.

Discharge Demand Forecast

Visualizes discharge forecasting trends.

The dashboard improves stakeholder understanding of forecasting results.

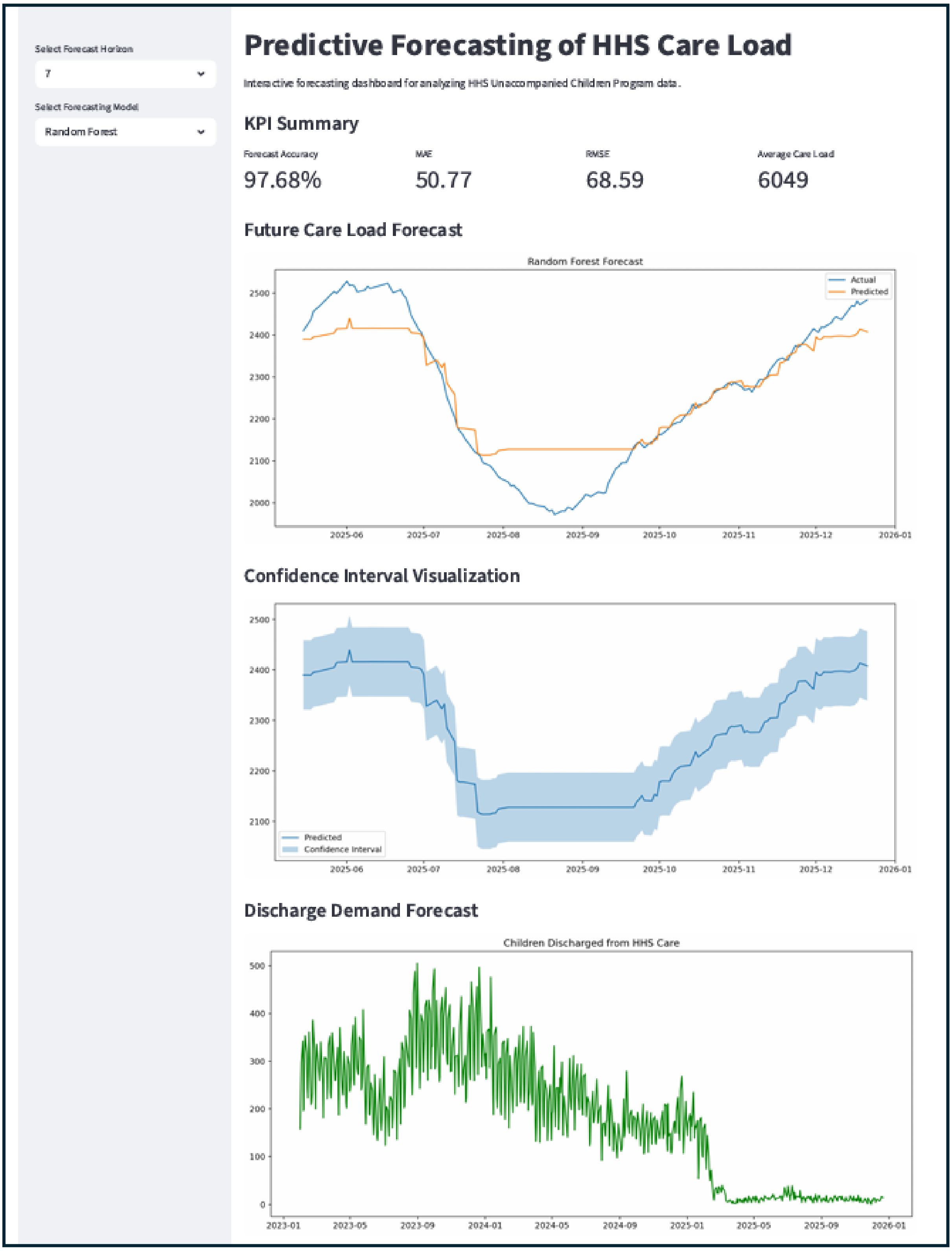

Figure 8: **Interactive Streamlit Forecasting Dashboard**

RESULTS AND INSIGHTS

The forecasting models successfully captured operational forecasting trends and care load behavior.

Key findings include:

Historical Fluctuations

Care load trends showed periodic operational fluctuations over time.

Machine Learning Performance

Random Forest and Gradient Boosting produced strong forecasting accuracy.

Statistical Forecasting

ARIMA successfully captured time-series forecasting behavior.

Dashboard Insights

Interactive visualization improved accessibility and forecasting interpretation.

Forecasting Comparison

Machine learning forecasting models demonstrated improved performance compared with simpler statistical approaches.

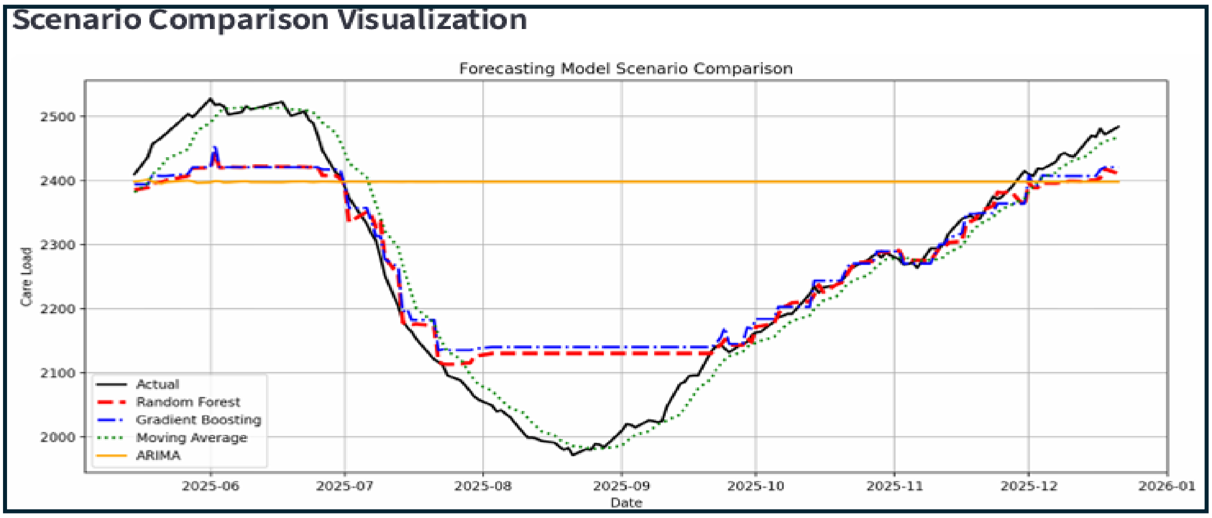

Figure 9: **Forecasting Scenario Comparison Visualization**

CONFIDENCE INTERVAL ANALYSIS

Confidence interval forecasting visualization was used to analyse forecasting uncertainty and prediction reliability.

The shaded confidence interval region represents the expected range of forecasting variation and helps evaluate operational forecasting stability.

Confidence interval analysis improves understanding of forecasting reliability and supports operational decision-making.

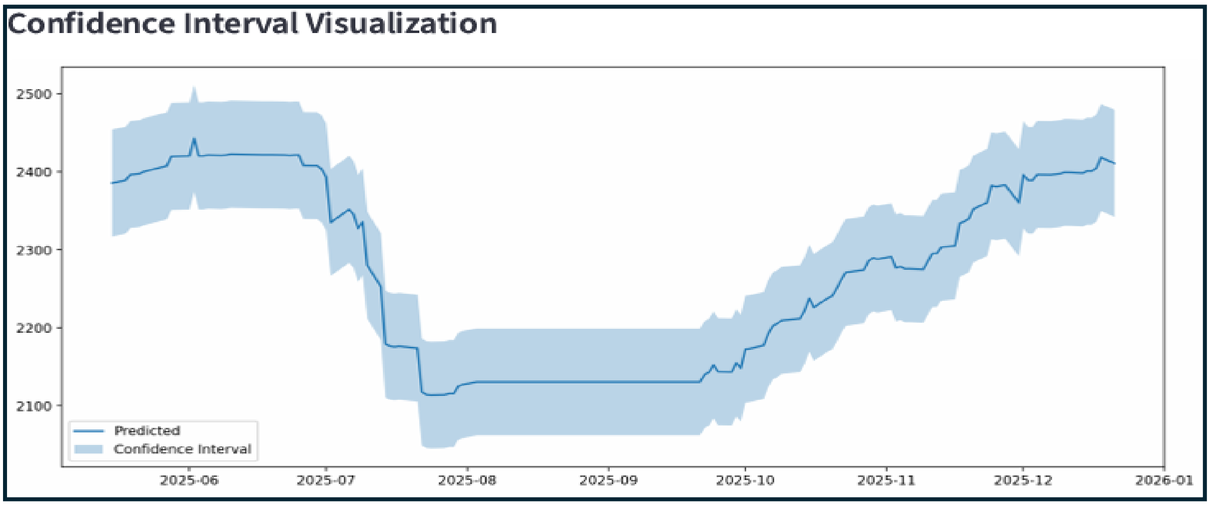

Figure 10: **Confidence Interval Forecast Visualization**

RECOMMENDATIONS

Based on forecasting analysis, several recommendations are proposed.

Real-Time Forecasting

Organizations should integrate real-time forecasting systems.

Better Data Collection

Operational monitoring systems should improve data quality and consistency.

Automated Dashboards

Interactive dashboards should be used for operational forecasting monitoring.

Predictive Decision Making

Forecasting analytics should support policy-level operational planning.

Advanced Forecasting Systems

Future forecasting models can include deep learning approaches.

CONCLUSION

This project successfully developed a predictive forecasting analytics solution for the HHS Unaccompanied Children Program.

The implementation included:

- data preprocessing
- exploratory data analysis
- forecasting model development
- evaluation metrics
- interactive dashboard visualization

Both statistical forecasting and machine learning forecasting approaches were implemented and compared.

The project demonstrated how predictive analytics can improve operational forecasting, resource planning, and decision-making processes.

The Streamlit dashboard further enhanced accessibility of forecasting insights through interactive analytical visualization.

FUTURE SCOPE

Future improvements for this project may include:

- cloud deployment
- real-time API integration
- deep learning forecasting
- automated alert systems
- advanced visualization dashboards
- live operational forecasting

Future forecasting systems can further improve operational planning efficiency and predictive analytics capabilities.

REFERENCES

The project utilized learning resources and documentation from:

- Python documentation
- Pandas documentation
- Scikit-learn documentation
- Streamlit documentation
- Statsmodels documentation
- machine learning forecasting tutorials
- time-series forecasting resources

These resources helped implement forecasting models and dashboard development successfully.